CM3121 – Advanced Organic Chemistry

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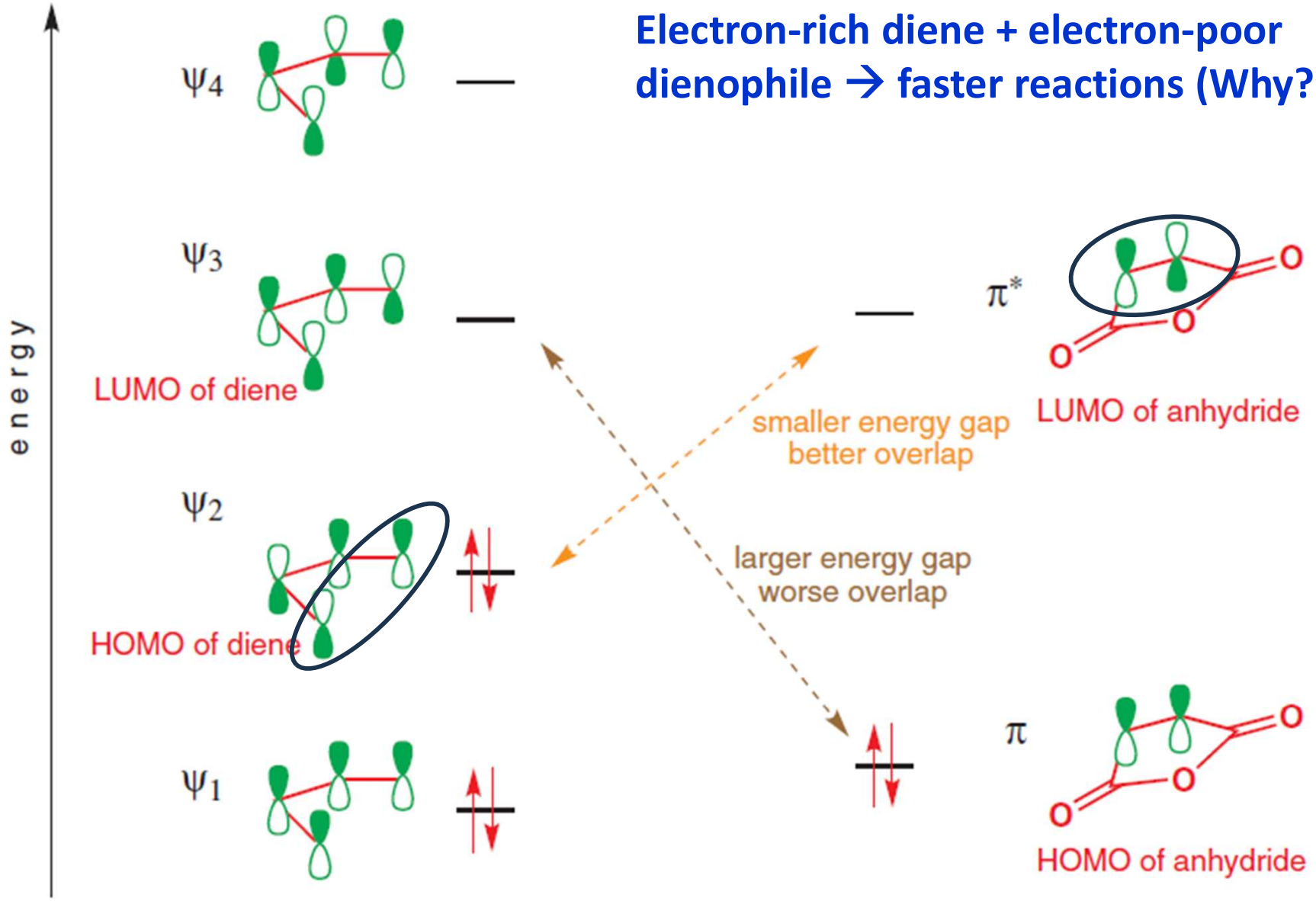
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Cycloadditions (part 2)

Overview

- Woodward–Hoffmann rules
- Reaction rates of Diels–Alder reactions—FMO analysis
- Regioselectivity of Diels–Alder reactions
- Catalysis in Diels–Alder reactions
- [6+4] and [4+3] cycloaddition
- [2+2] cycloaddition

The frontier orbitals of DA reaction



Electron-rich diene + electron-poor dienophile → faster reactions (Why?)

6. Woodward–Hoffmann rules

- **In a thermal pericyclic reaction, the total number of (4q + 2)*s* and (4r)*a* components must be odd (q and r are whole numbers).**

In an allowed thermal pericyclic reaction this sum:

number of suprafacial components with 2, 6, or 10 electrons
\+ number of antarafacial components with 0, 4, or 8 electrons
= an odd number

The Nobel Prize in Chemistry 1981



Photo from the Nobel Foundation archive.
Kenichi Fukui
Prize share: 1/2

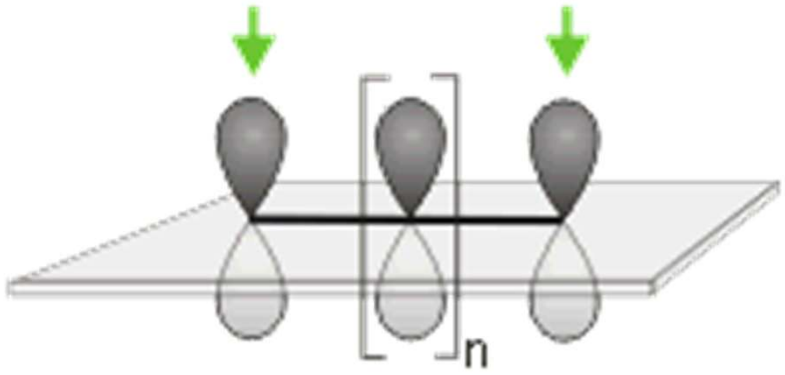


Photo from the Nobel Foundation archive.
Roald Hoffmann
Prize share: 1/2

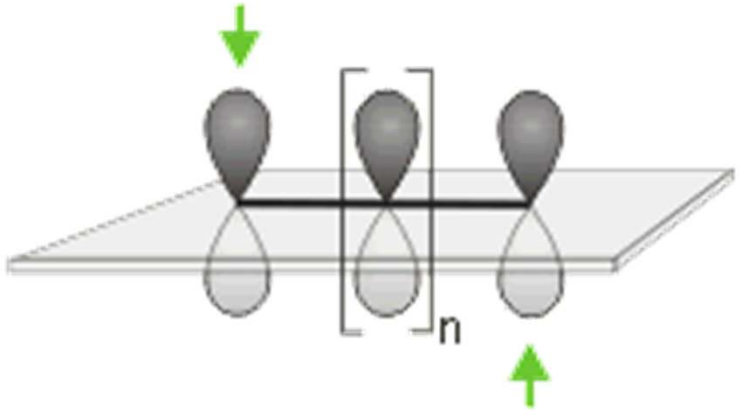
The Nobel Prize in Chemistry 1981 was awarded jointly to Kenichi Fukui and Roald Hoffmann "for their theories, developed independently, concerning the course of chemical reactions."

'Allowed' doesn't mean that it will happen.
It just means it is theoretically possible.

Woodward–Hoffmann rules



Suprafacial

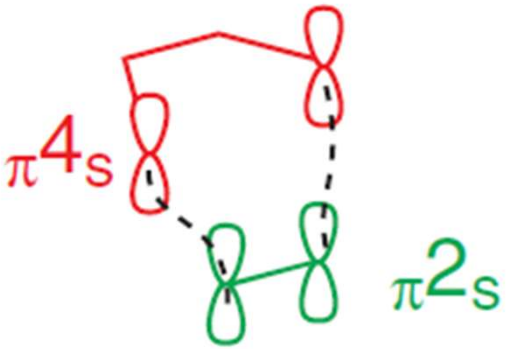


Antarafacial

A **suprafacial** component forms new bonds on the same face at both ends.

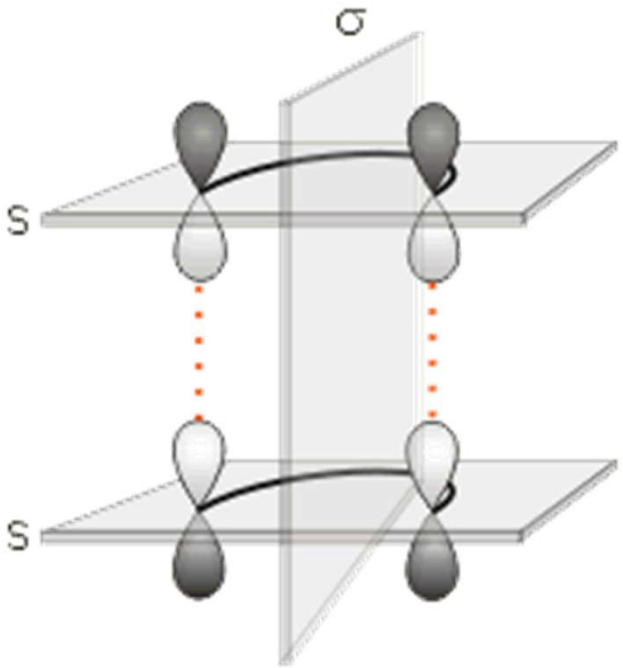
An **antarafacial** component forms new bonds on opposite faces at both ends.

Woodward–Hoffmann rules



DA reactions:

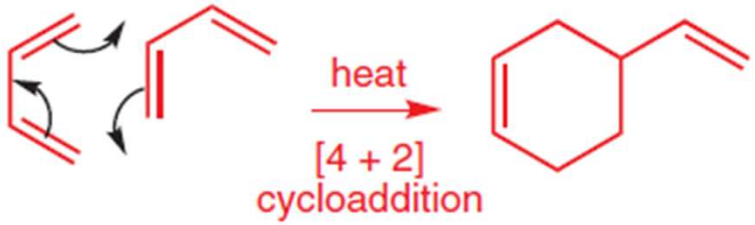
- **1 * (4q + 2)s component (the alkene)**
- **0 * (4r)a component.**
- **Total = 1 so it is an allowed reaction.**

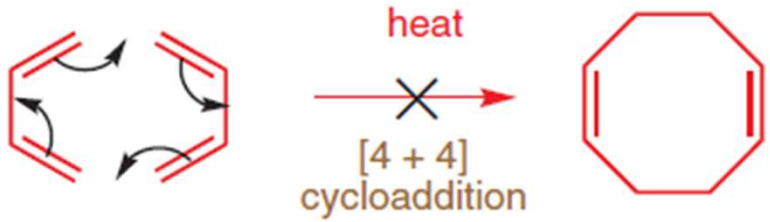


suprafacial-suprafacial approach

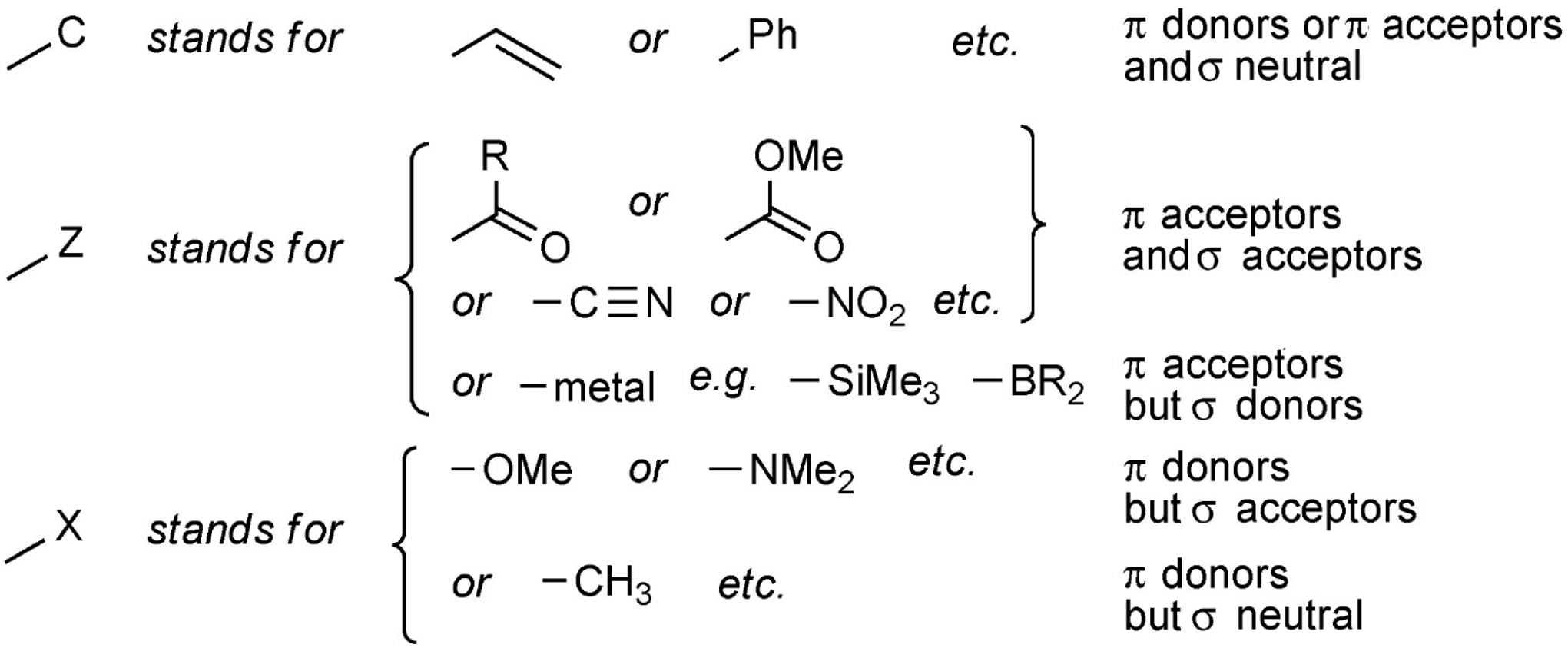
Practice

Use Woodward–Hoffmann rules to explain: 1,3-butadiene undergoes thermal [4+2] reaction, but not thermal [4+4] reaction, assuming suprafacial-suprafacial approach





7. Substituents classifications

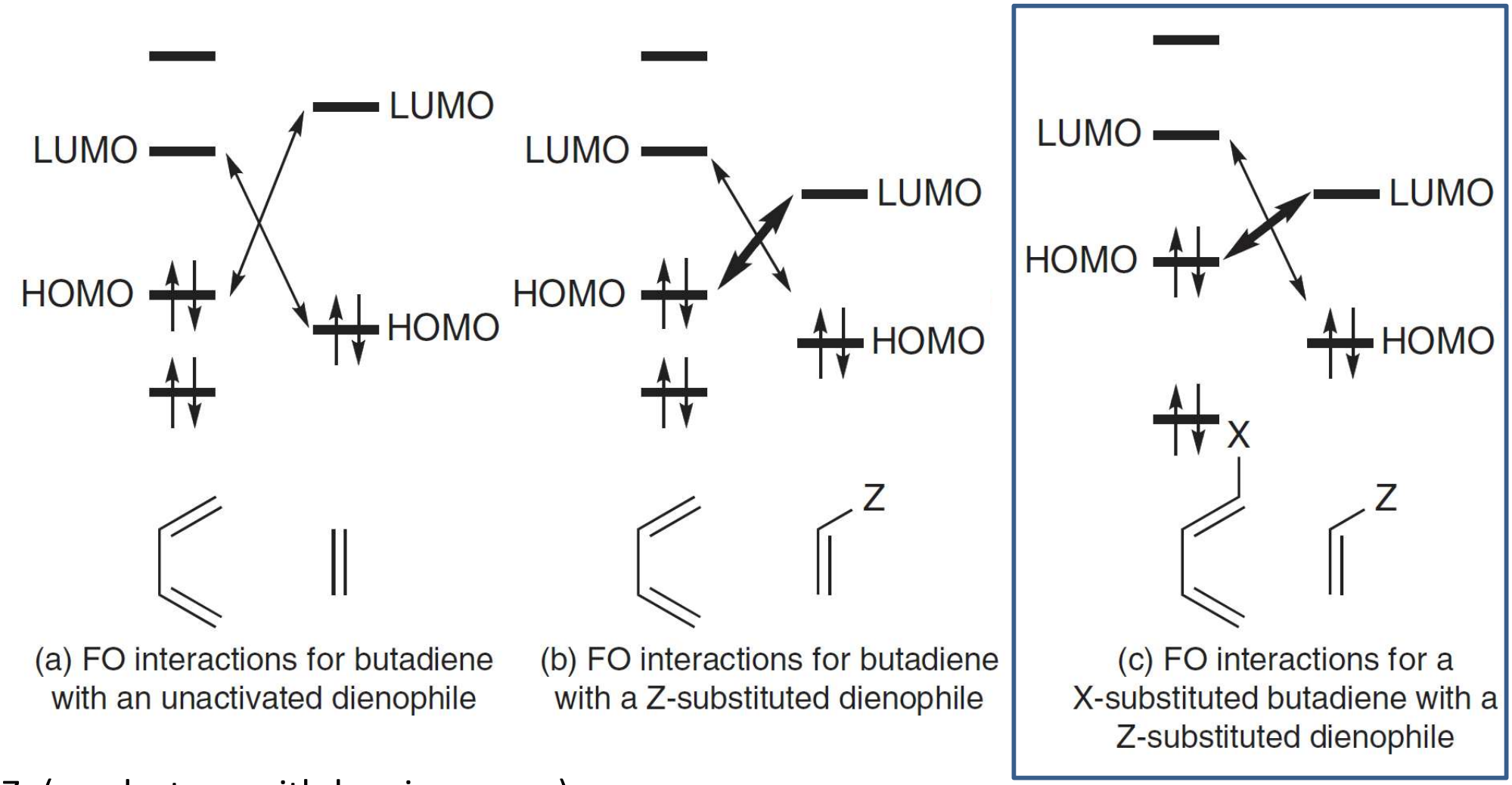


Definitions and character of C-, Z- and X- type substituents

The effects of the three kinds of substituent on the molecular orbital

- C- (extra conjugation)
  - raises the energy of the HOMO
  - lowers the energy of the LUMO
- Z- (an electron-withdrawing group)
  - slightly lowers the energy of the HOMO
  - substantially lowers the energy of the LUMO
- X- (an electron-donating group)
  - substantially raises the energy of the HOMO
  - slightly raises the energy of the LUMO

High reactivity of electron rich diene and electron deficient dienophile



(a) FO interactions for butadiene with an unactivated dienophile

(b) FO interactions for butadiene with a Z-substituted dienophile

(c) FO interactions for a X-substituted butadiene with a Z-substituted dienophile

Z- (an electron-withdrawing group)
X- (an electron-donating group)

small LUMO(dienophile)-HOMO(diene) gap leads to faster reaction.

8. Regioselectivity in Diels–Alder reactions



- The polarization of the π electrons determine the regioselectivity

  "Electrophilic" position of the dienophile reacts with the "nucleophilic" position of the diene.

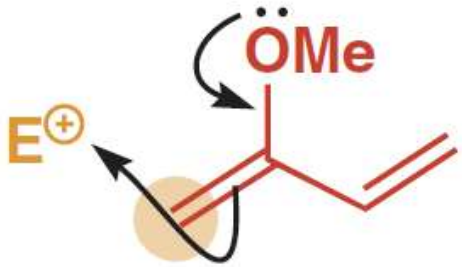
Regioselectivity in Diels–Alder reactions (C2-substituted diene)

C2-substituted diene:



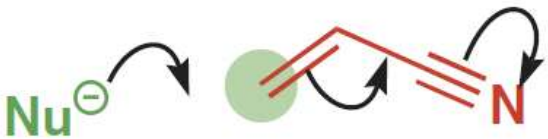
Step 1

reaction of the diene with an electrophile



Step 2

reaction of the dienophile with a nucleophile



Step 3

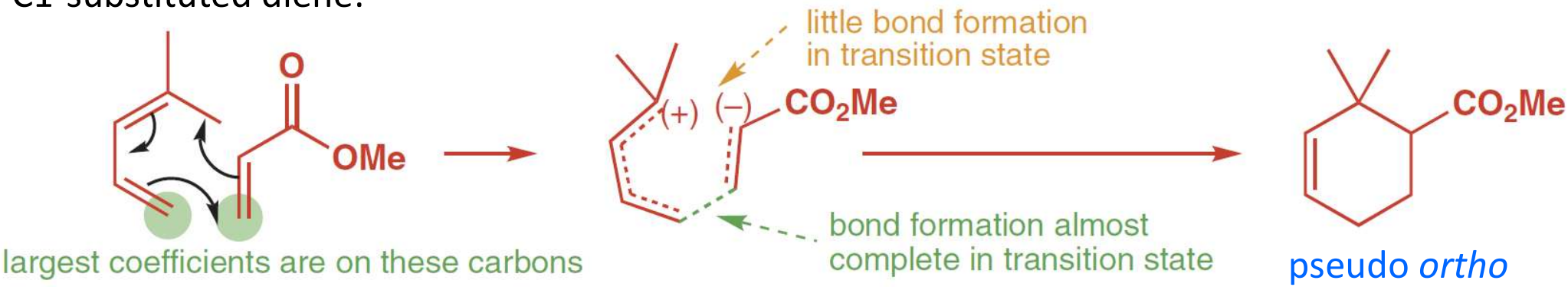


pseudo *para*

Regioselectivity in Diels–Alder reactions (C1-substituted diene)

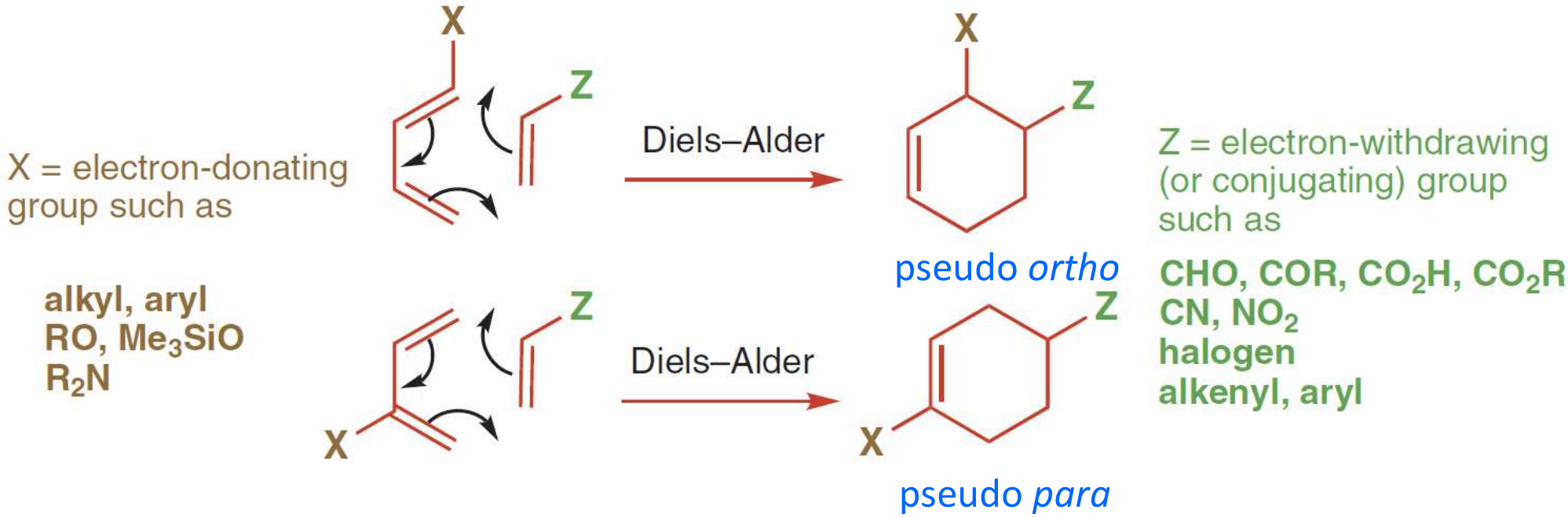
Concerted, but unsymmetrical transition state

C1-substituted diene:



Regioselectivity in Diels–Alder reactions

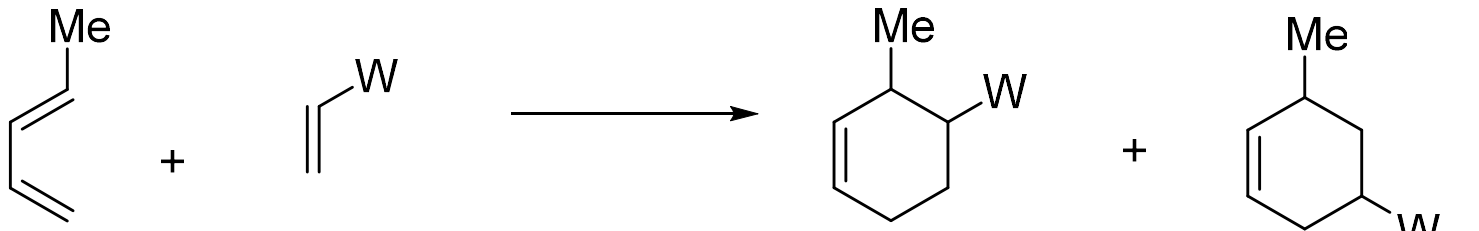
Top: C1-substituted diene



Bottom: C2-substituted diene

Regioselectivity in Diels–Alder reactions (examples)

C1-substituted diene:



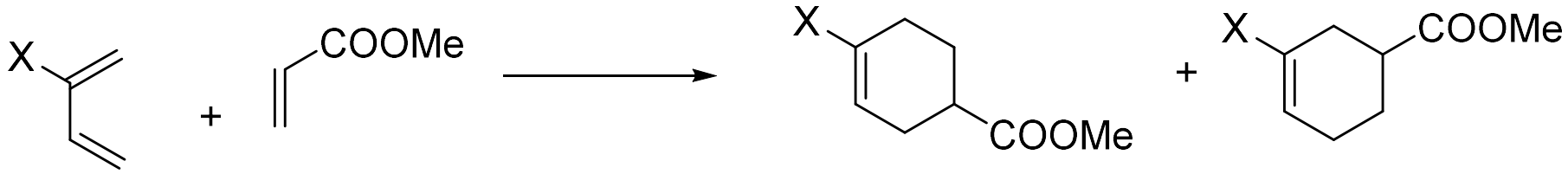
di-substituted cyclohexenes

"ortho" (1,2) "meta" (1,3)

W = CN, 100 °C — 9: 1

W = COOR, 200 °C — 6.8: 1

C2-substituted diene:

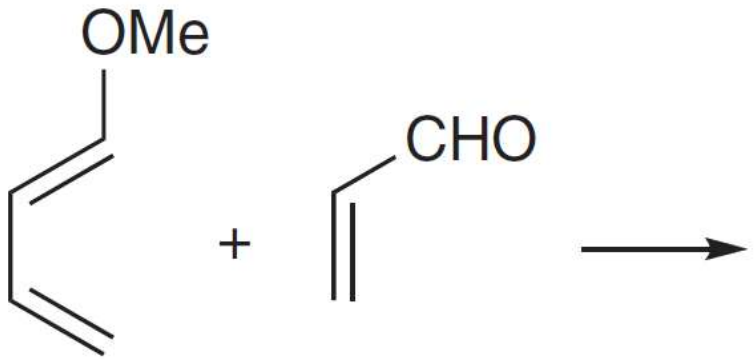


"para" (1,4) "meta" (1,3)

X = Me — 6: 1

X = OMe — 10: 1

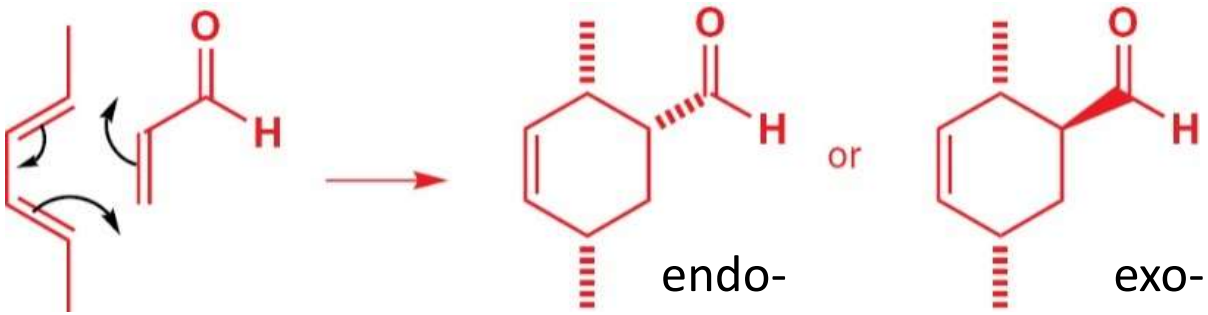
9. Combination of Regioselectivity and Stereoselectivity in Diels–Alder reactions



Predict the stereochemistry of cycloaddition product

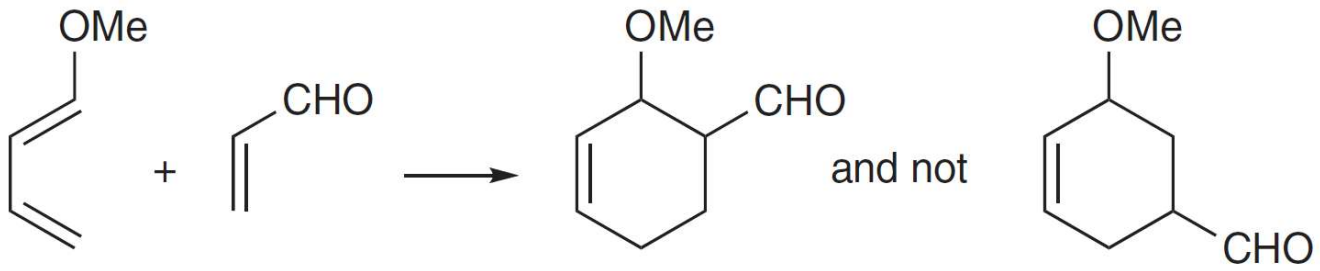
- Step 1: regioselectivtiy (ortho- for C1-substituted diene)
- Step 2: stereoselectivty (endo-)

Recall:



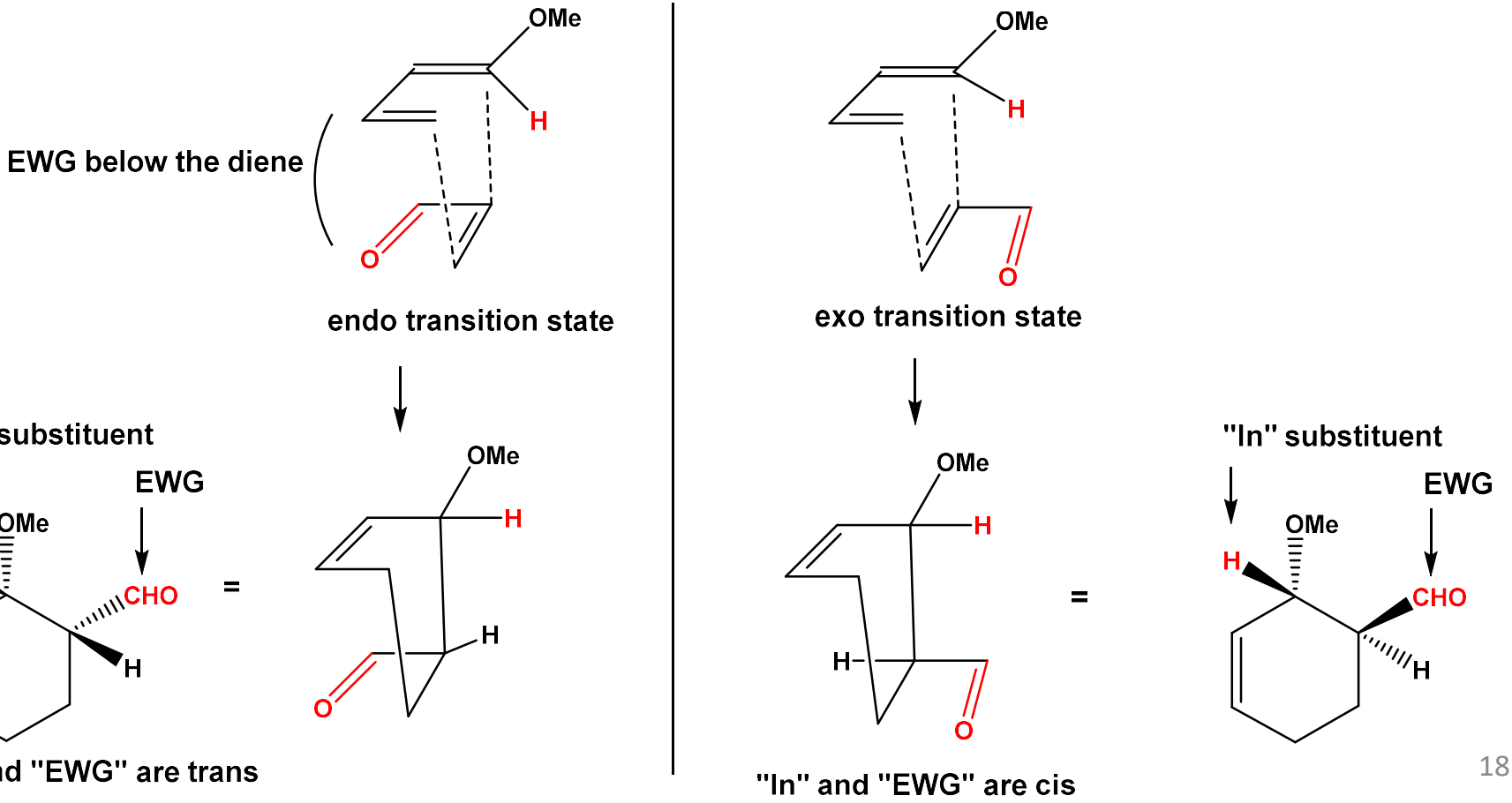
endo- exo-

- Step 1: regioselectivtiy (ortho- for C1-substituted diene)

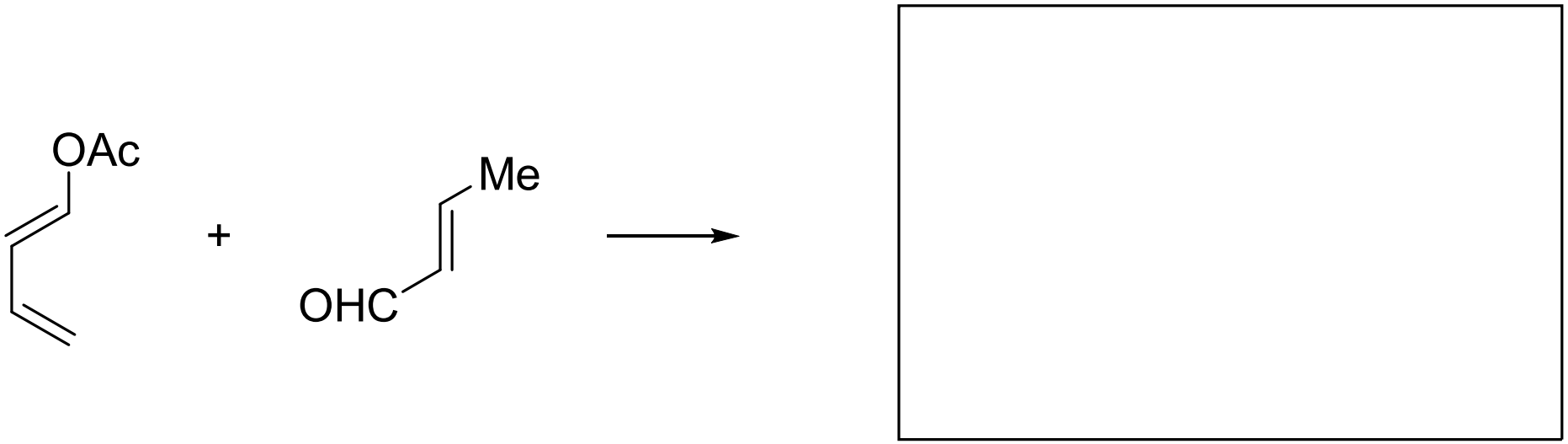


Cis- or trans-?

- Step 2: stereoselectivty (endo-)

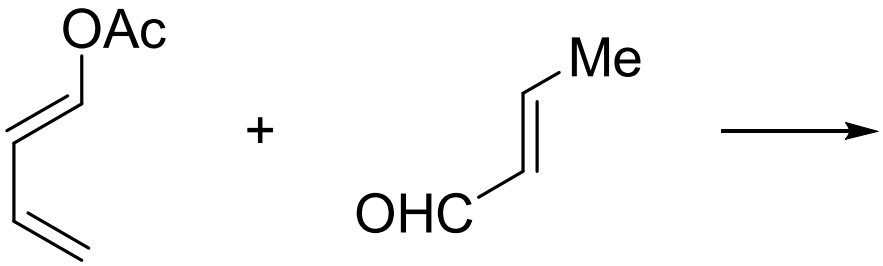


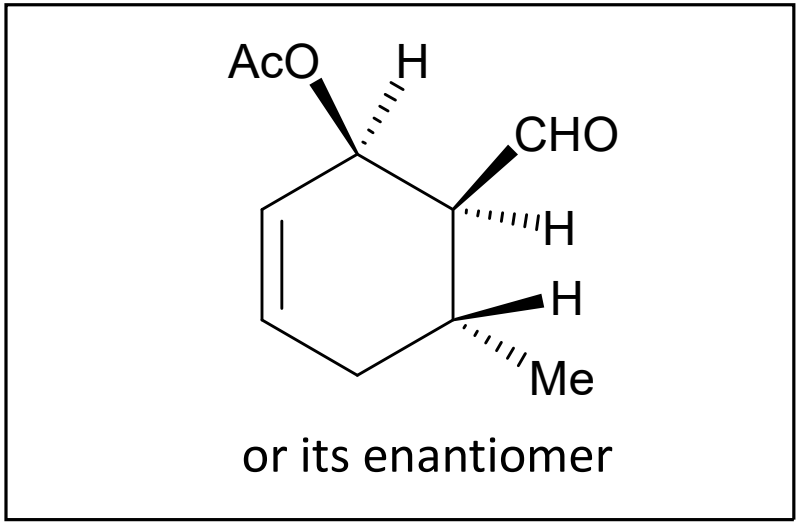
Practice

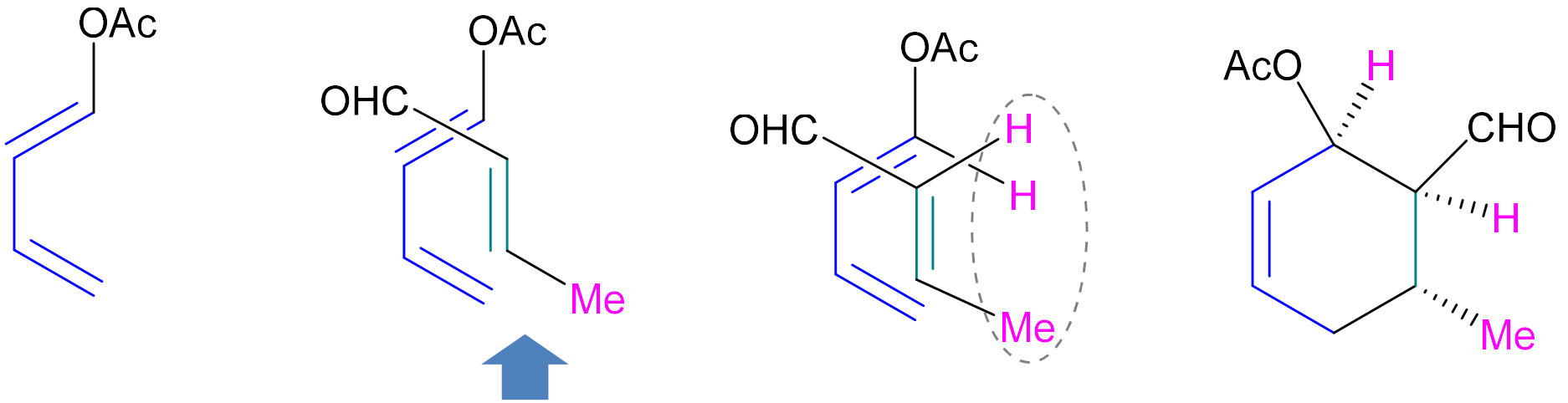


- Step 1: regioselectivtiy (ortho-)
- Step 2: stereoselectivty (endo-)
- Step 3: stereospecificity

Practice



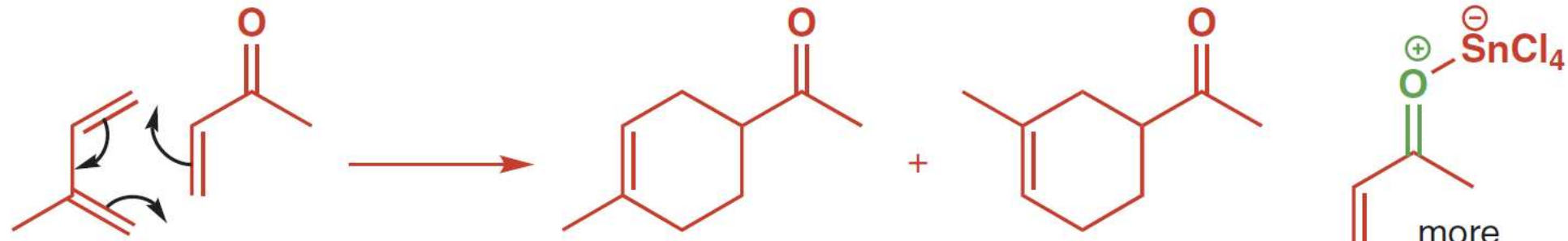




- Step 1: regioselectivtiy (ortho-/para-)
- Step 2: stereoselectivty (endo-)
- Step 3: stereospecificity

10. Lewis acid catalysis in Diels–Alder reactions

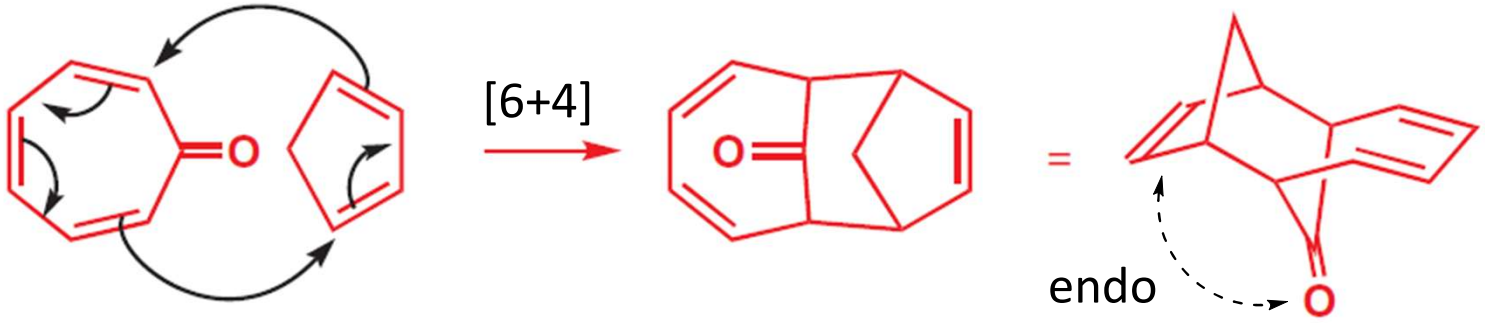
- Reactivity (LUMO energy)
- Regioselectivity (polarization of π bond)

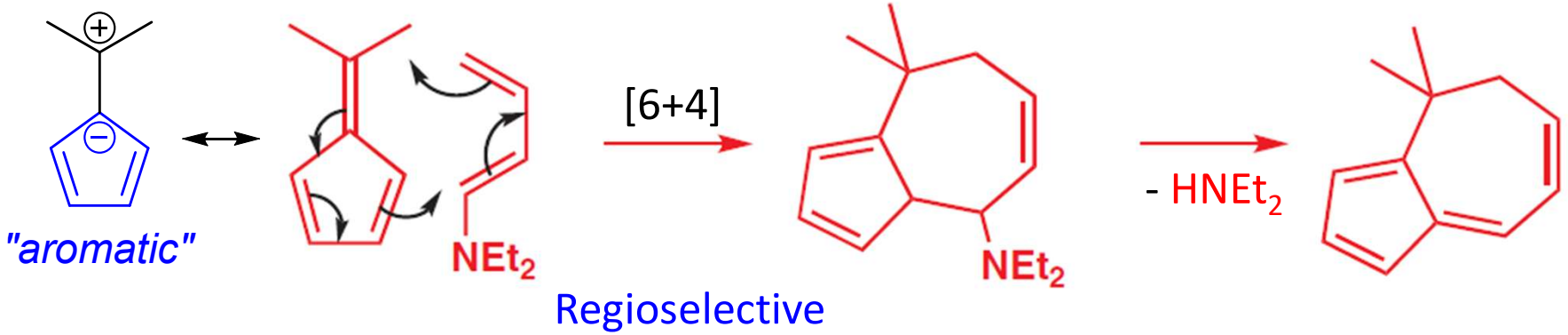


11. Beyond formation of 6-membered rings [6+4] cycloaddition

Is this type of reactions allowed according to Woodward–Hoffman rules?

Woodward-Hoffmann rules: One component is (4q+2)s, the other component is (4r)s, where q and r are integers (0, 1, 2…).

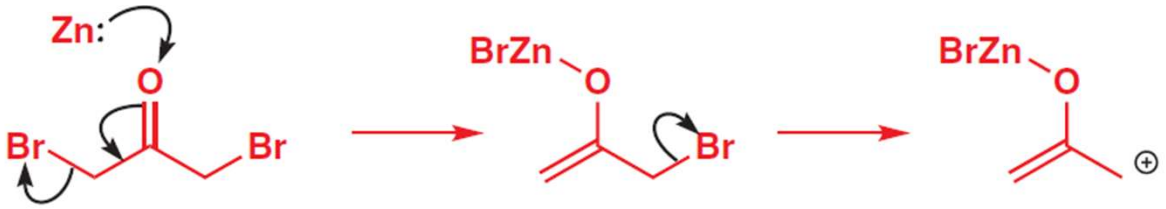




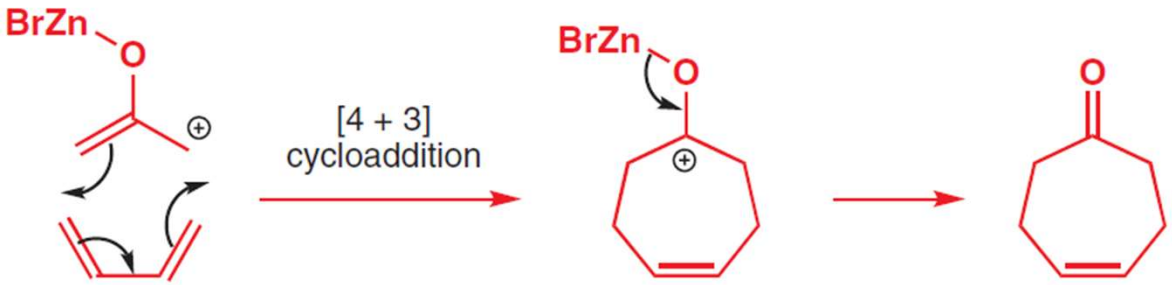
Regioselective

In other words: Thermal pericyclic cycloadditions are allowed if the total number of electrons involved can be expressed in the form of **4n+2**, where n is an integer (suprafacial-suprafacial).

12. [4+3] cycloaddition

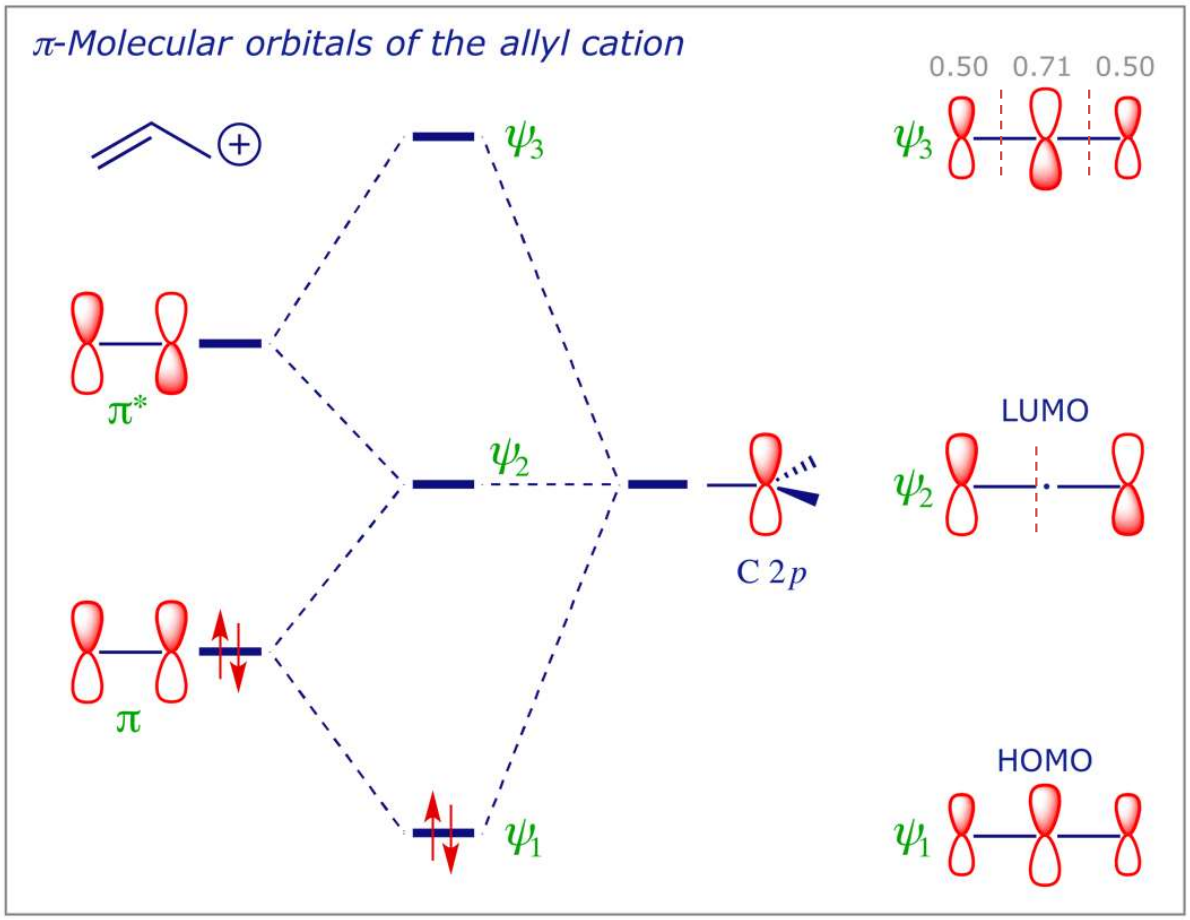


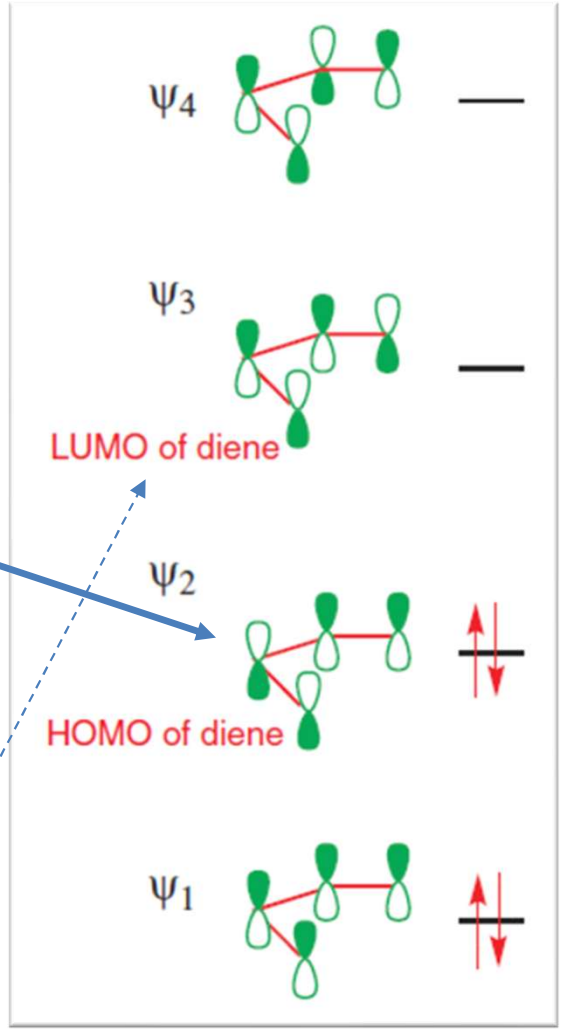
The allyl cation has three atoms but only two electrons

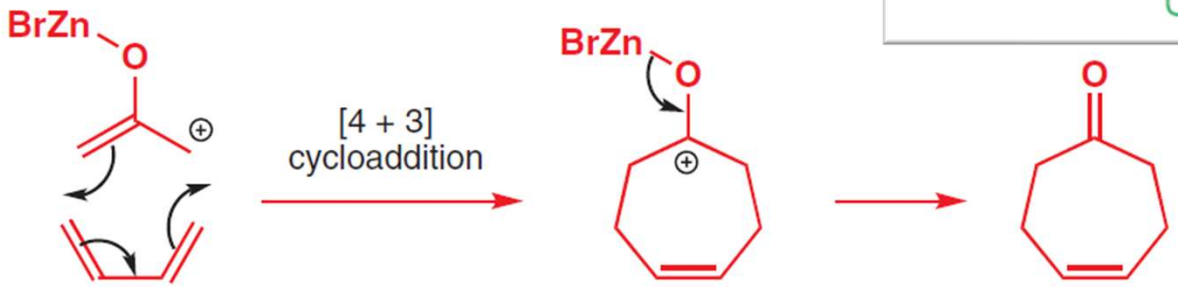


Draw the frontier molecular orbitals of this reaction. Is it allowed?

- [4+3] cycloaddition

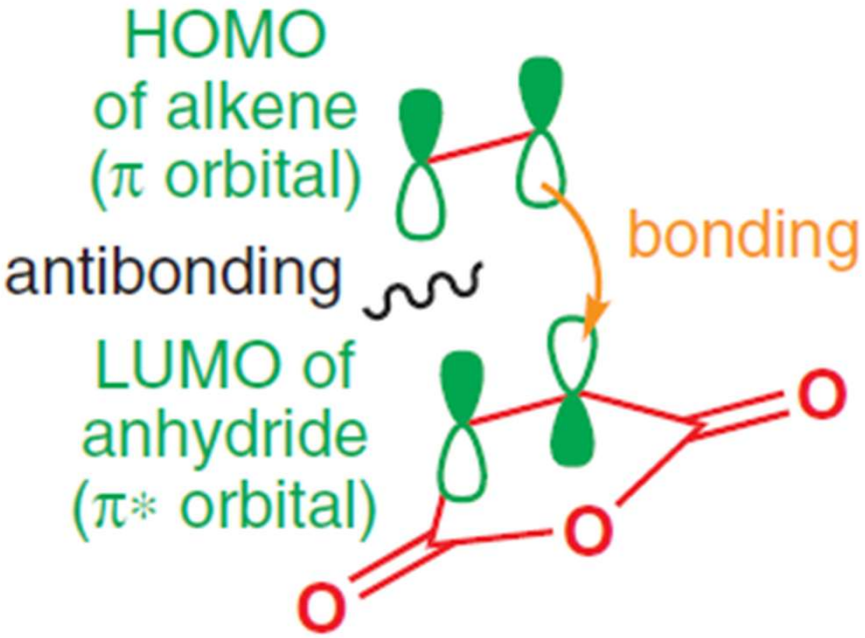




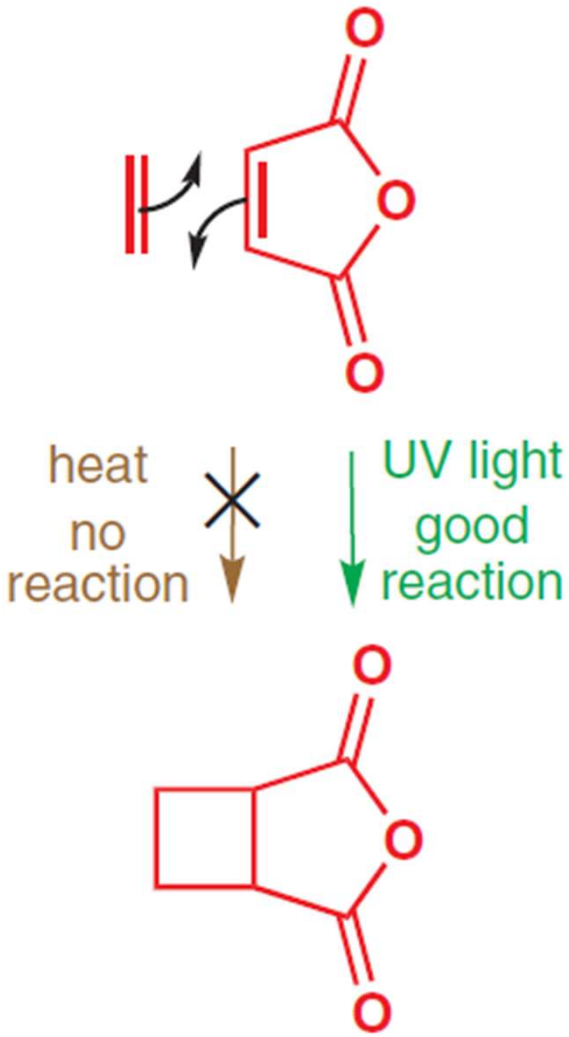


Draw the frontier molecular orbitals of this reaction. Is it allowed?

13. Photochemical [2 + 2] cycloadditions



HOMO/LUMO interactions are out of phase (forbidden)



Woodward–Hoffmann rules

- **In a thermal pericyclic reaction the total number of (4q + 2)s and (4r)a components must be odd.**

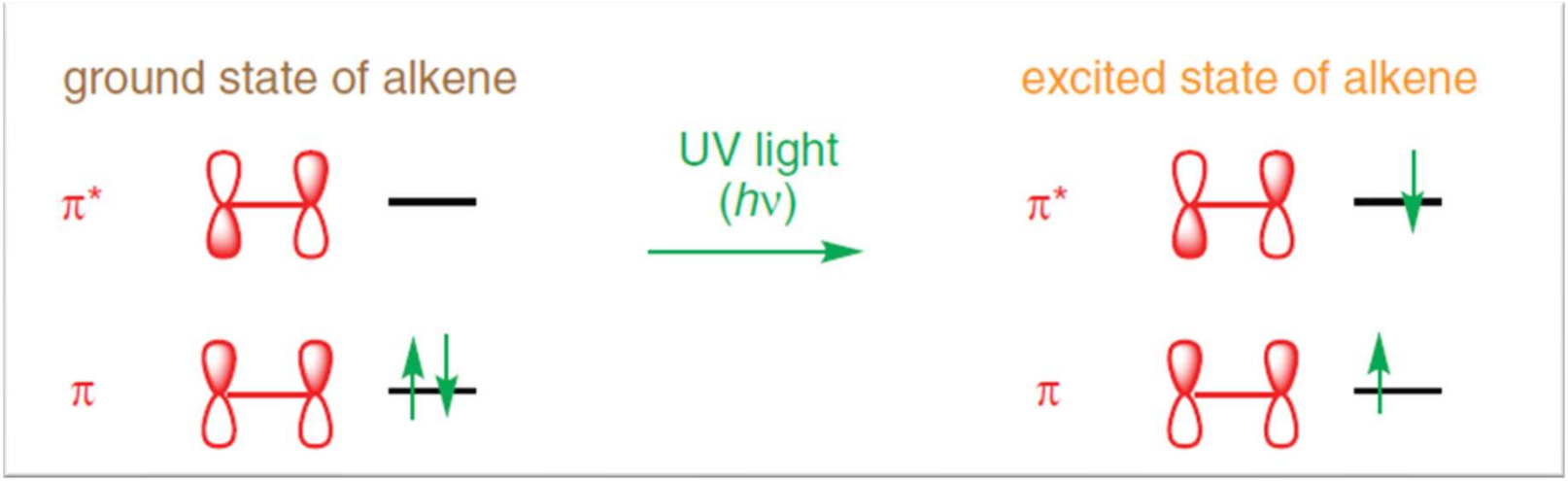
In an allowed thermal pericyclic reaction this sum:

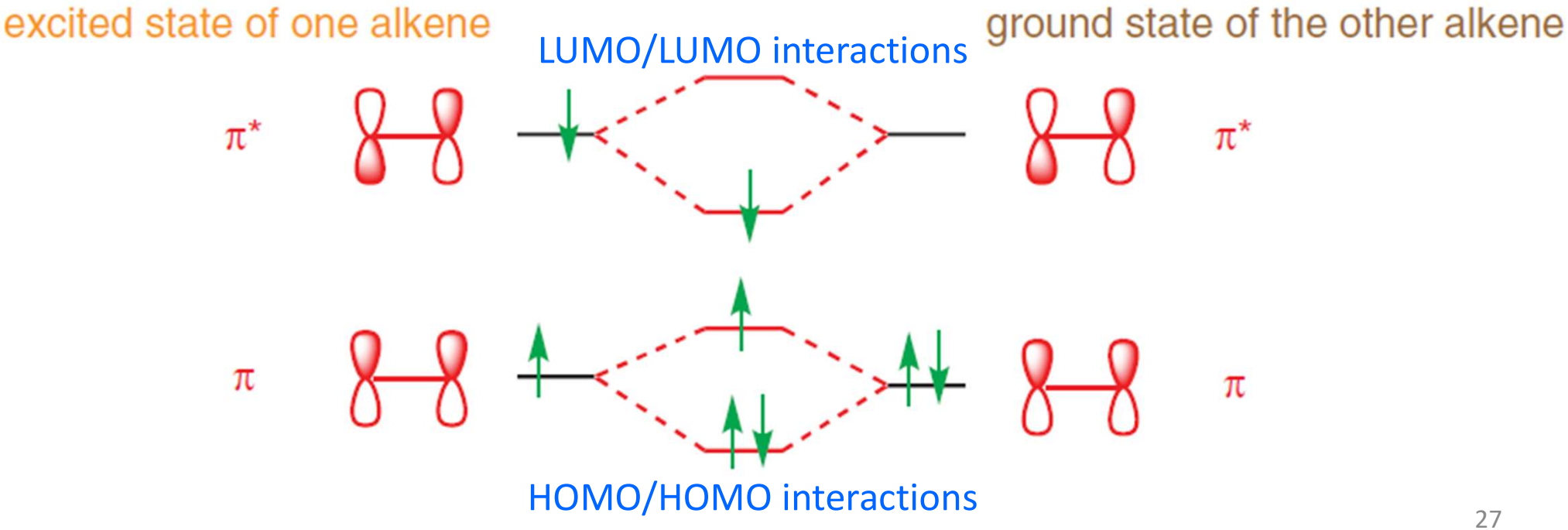
number of suprafacial components with 2, 6, or 10 electrons
\+ number of antarafacial components with 0, 4, or 8 electrons
= an odd number

Is [2+2] reaction allowed according to Woodward–Hoffman rules?

‘Allowed’ doesn’t mean that it will happen.
It just means it is theoretically possible.

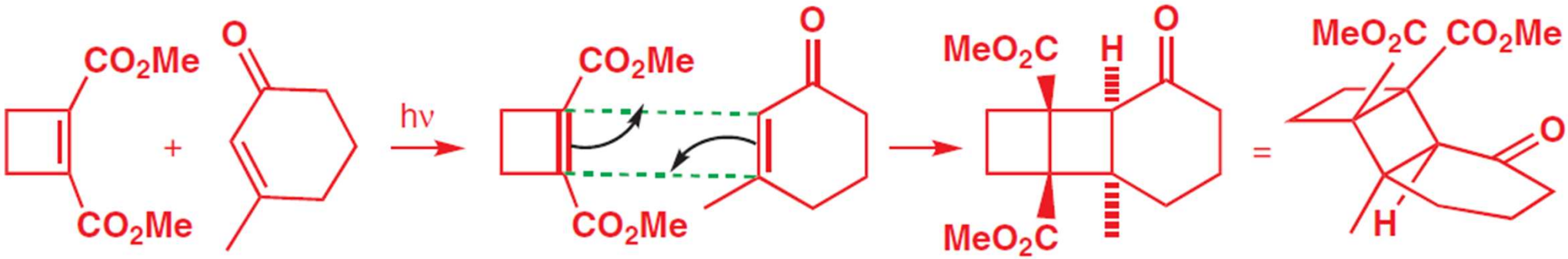
Photochemical [2 + 2] cycloadditions





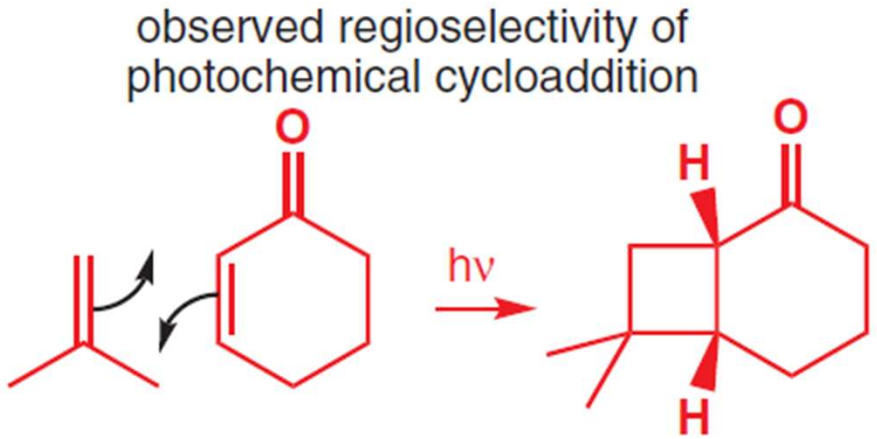
Photochemical [2 + 2] cycloadditions

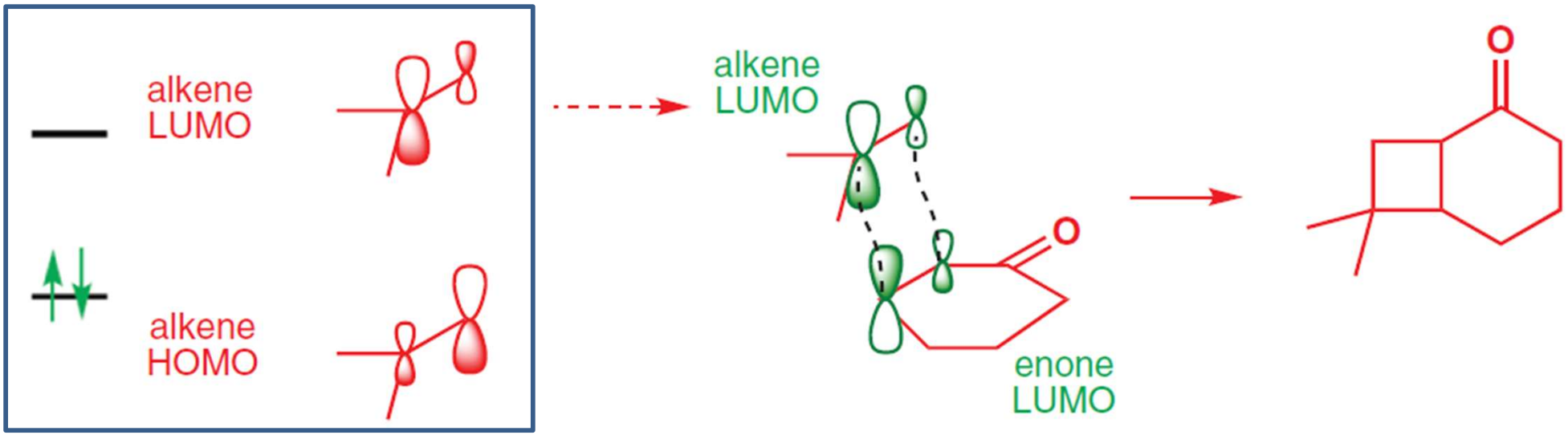
- Stereospecific within each component
- Stereoselective: the least sterically hindered transition state (no endo selectivity)



Photochemical [2 + 2] cycloadditions

- Regioselectivity



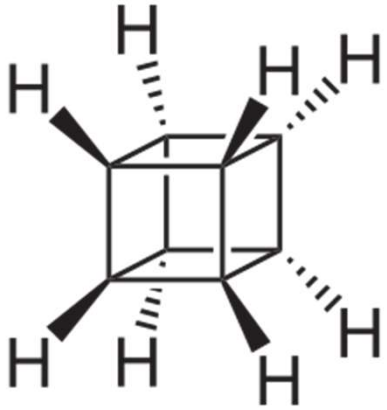


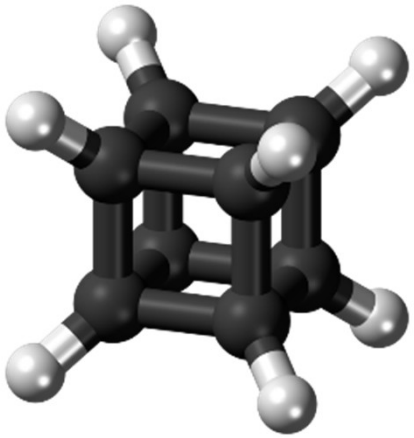
HOMO/HOMO and LUMO/LUMO interactions under photochemical conditions

Cubanes

Cubane

C_8H_8





First synthesized in 1964 by Philip Eaton and Thomas Cole

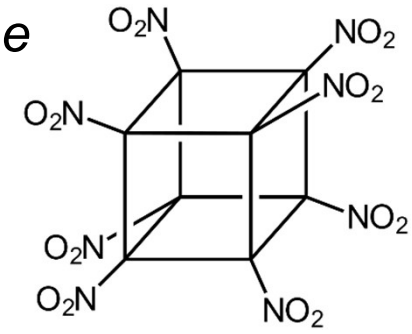
strained chemical bonds

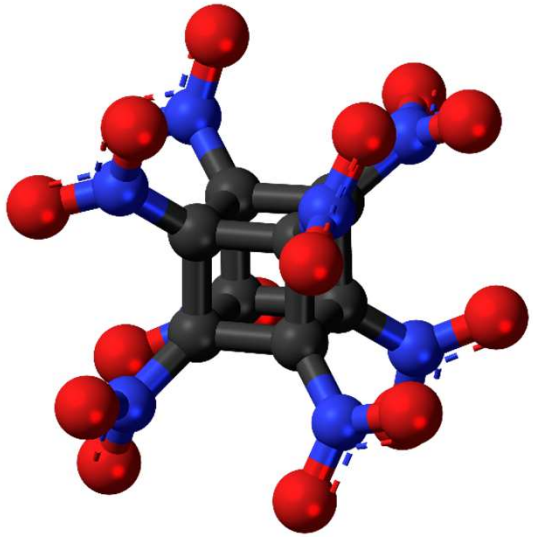
Octanitrocubane

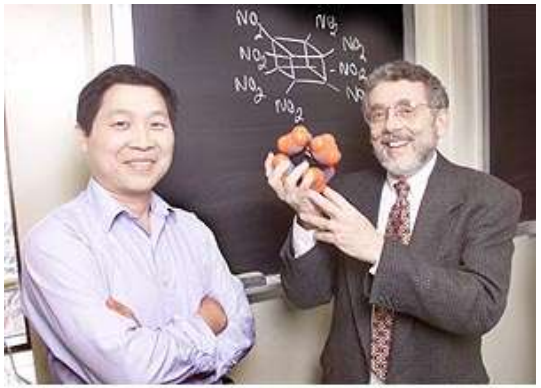
$C_8(NO_2)_8$



CO_2 and N_2



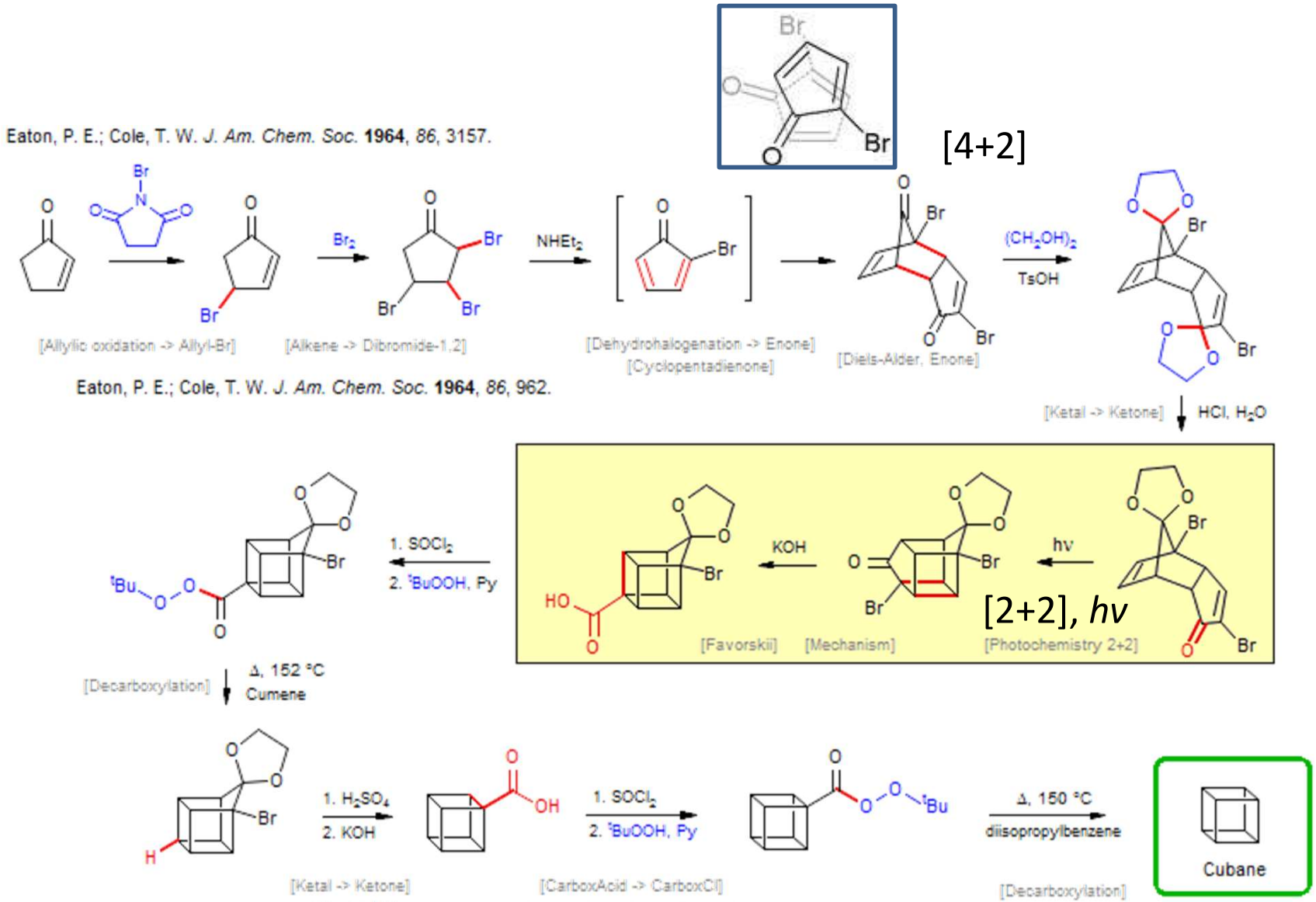




First synthesized by Philip Eaton and Mao-Xi Zhang in 1999

The fastest and most effective chemical explosive (also shock-insensitive).

Synthesis of Cubane



Summary

Diels Alder reactions:

- Electron-rich diene + electron-poor dienophile → faster reactions
- Stereospecific (1,4-di-substituted diene or 1,2-di-substituted dienophile)
- Stereoselective: it favors endo-transition state (combination of substituted diene and substituted dienophile)
- Regioselective: it favors “para-” / “ortho-” products (electron-rich diene + electron-poor dienophile)

Formation of various cyclic compounds through various cycloaddition reactions:

- Thermal [6+4] and [4+3] cycloadditions
- Photochemical [2 + 2] cycloadditions
- Thermal [2 + 2] cycloadditions (cumulated double bonds)